VCE-BRT-001
Revision 1.0
2025

Builder's *Report*

VOLTA CONFLUENCE ENGINE — DOCUMENT YOUR BUILD FOR EVERY BUILDER WHO FOLLOWS

INSTRUMENT	LICENCE	SUBMIT TO
Volta Confluence Engine	CC BY-NC	mumblejinx.github.io/opensource

WHY THIS REPORT MATTERS

The Volta Confluence Engine has never been physically built. Every confidence figure, every pass threshold, every machining parameter in the builder package is an engineering estimate — not a measured result. Your build is the first physical test of this design. The numbers you measure here will move specifications from ESTIMATED to CONFIRMED, and the problems you encounter will save every builder who follows you from repeating them. This report is not administrative paperwork. It is the project's primary mechanism for becoming more honest over time.

Fill in what you can. Partial reports are valuable. A report that says "I built the energy system, here are my measured voltages, I stopped before completing the signal chain" is genuinely useful. Submit whatever you have.

SECTION 1

Builder *Identity*

You may submit this report anonymously. If you provide contact details, the project may follow up with questions about your build — only with your permission.

BUILDER NAME OR HANDLE

Real name or pseudonym — your choice

CONTACT (OPTIONAL)

Email or website — only if you want to be co

LOCATION (COUNTRY / REGION)

e.g. Germany, Bavaria

Useful for understanding build conditions and supplier availability by region.

BUILD START DATE

Month and year

REPORT SUBMISSION DATE

Month and year

BUILD STATUS AT SUBMISSION

e.g. Complete · Signal chain only · Mechanic

BUILDER BACKGROUND

Briefly describe your relevant background — e.g. professional machinist, amateur electronics, clockmaking experience, no prior experience. This helps contextualise your observations.

SECTION 2

Build *Configuration*

Which options did you build?

- [ ] Option A — Standard Hand Crank
- [ ] Option B — Gear-Assisted Hand Crank
- [ ] Option B+ — Peripheral Light Indicator
- [ ] Option C — Gravity Weight Tower
- [ ] Option D — Audience Crank Unit

SUT (STEP-UP TRANSFORMER) USED

e.g. Lundahl LL1931 1:10 · Jensen JT-346 · Custom wound

The SUT specification is the most underspecified element in the current document. Your choice is valuable data.

CARTRIDGE(S) USED

Make, model, and output voltage

CLOCKMAKER / SPRING MOTOR MANUFACTURER

Name and country — with permission to share

Other builders need this. The clockmaker is the longest lead time and hardest-to-find commission in the build.

FIELD COIL SPEAKER BUILDER

Name and country — with permission to share

CABINETMAKER

Name and country — with permission to share

TOTAL BUILD TIME (APPROXIMATE)

e.g. 380 hours over 9 months

TOTAL BUILD COST (APPROXIMATE, USD OR LOCAL CURRENCY)

e.g. $41,000 USD · €38,000

The cost estimate in the documentation is widely ranged. Your actual figure helps future builders budget realistically.

SECTION 3

Measured *Results*

Record your measured values against the document estimates. Leave blank if you did not measure that parameter. Circle or note where your measured value differs significantly from the document estimate — these are the most important data points.

Energy System

PARAMETER	DOCUMENT ESTIMATE	YOUR MEASURED VALUE	NOTES
`Generator output at comfortable crank rate`	20–28V AC RMS		
`C-W multiplier output (no load)`	~600V DC		
`C-W multiplier output (under load)`	540–660V DC		

Time to charge bank from empty (comfortable crank)	3–8 minutes		
Stored energy (measured by discharge test)	3,500–5,500J (estimated)		
Bank self-discharge rate	<5% per hour		
Bleed resistor discharge time to <50V	~60 seconds		

Turntable Mechanics

PARAMETER	DOCUMENT ESTIMATE	YOUR MEASURED VALUE	NOTES
Platter float height	1.2–1.8mm		
Bearing field at cartridge position	<5μT		
Mu-metal layers required	1 (estimated)		
Spring motor run time per full wind	25–28 minutes (estimated)		
Spring motor torque variation over run	<2% (estimated)		
Wow and flutter — with PLL	<0.05% WRMS (estimated)		
Wow and flutter — tuning fork (if used)	<0.15% WRMS (estimated)		
Isolation platform resonant frequency	1.5Hz target		

Signal Chain

PARAMETER	DOCUMENT ESTIMATE	YOUR MEASURED VALUE	NOTES

Heater voltage	6.3V DC ±5%		
Anode voltage	150–200V DC		
Quiescent current per stage	1.0–1.5mA		
Total gain at 1kHz	~52dB		
RIAA response deviation	±0.5dB target		
Channel separation at 1kHz	>60dB		
Noise floor (subjective)	Below vinyl surface noise		
Crossfader level at centre position	−3dB ±1dB		
Crossfader channel isolation	>50dB		

Field Coil Speakers

PARAMETER	DOCUMENT ESTIMATE	YOUR MEASURED VALUE	NOTES
Field coil current — compression drivers	20–30mA DC		
Field coil current — woofer	20–30mA DC		
Hum from drivers (subjective)	Inaudible with input shorted		
Output SPL at 1m (full system)	100–112dB avg (estimated)		

Option D — Audience Crank Unit (if built)

PARAMETER	DOCUMENT ESTIMATE	YOUR MEASURED VALUE	NOTES

Audience unit generator output	~24V AC		
Energy per 60-second crank session	~510J (estimated)		
Bank voltage increase per session	~10–12% of full bank (estimated)		
Audience engagement (subjective)	N/A — first build data		

SECTION 4

Overall *Verdict*

Your honest assessment of the build package. This directly updates the project's confidence figures.

Did the design work?

✓ YES – BUILT AND PLAYS	◈ YES – WITH MODIFICATIONS	✕ NO – DID NOT FUNCTION

WHAT MODIFICATIONS DID YOU MAKE, AND WHY?

Describe any changes from the documented specification — different components, different dimensions, different approaches. Explain what drove the change. 'I used a Jensen SUT instead of Lundahl because it was available locally' is useful. 'I changed the C-W stage count from 12 to 14 because I could only reach 520V at comfortable crank

Your first-prototype confidence estimate

The document claims 88% first-prototype confidence. Having built it, what would you put that figure at?

YOUR FIRST-PROTOTYPE CONFIDENCE ESTIMATE (%)

e.g. 85% · 92% · Too early to say

YOUR SECOND-PROTOTYPE CONFIDENCE ESTIMATE (%)

Based on what you'd change on a second bui

SECTION 5

What the Documents *Got Right*

Specifications that held exactly as stated move from ESTIMATED or ROUGH DRAFT to CONFIRMED in the next document revision. Please be specific — "the signal chain worked" is not useful. "The RIAA component values from IEC 60098 produced ±0.3dB response accuracy without adjustment" is very useful.

List the specifications, dimensions, component values, and procedures that worked without modification. Note which certainty label each item carried in the original document — confirming an Estimated figure is more significant than confirming a Confirmed figure.

SECTION 6

What Needed *Changing*

This section is the most valuable part of the report. Every item here prevents the next builder from spending time on a problem you already solved.

SPECIFICATION ERRORS (DOCUMENT STATED SOMETHING INCORRECT)

e.g. 'The stated mainspring dimensions (25mm × 0.6mm × 1200mm) produced only 19 minutes run time, not the documented 25–28. The clockmaker recommended 1400mm active length for the target run time.' Include your corrected figure wherever possible.

GAPS (DOCUMENT DID NOT TELL YOU SOMETHING YOU NEEDED TO KNOW)

e.g. 'The C-W multiplier section does not specify how to calculate the number of stages from the generator output voltage. I had to derive this independently. Suggest adding: stages = target_voltage ÷ (peak_input × 2), rounded up.' Describe the gap and, if possible, how you filled it.

ROUGH DRAFT ITEMS YOU WERE ABLE TO CONFIRM

e.g. 'The lathe parameters for OFHC copper (600 RPM, HSS tool, no coolant) produced acceptable results with minor adjustment — I found 700 RPM gave a better surface finish on the facing operation. Suggest updating the document to 600–800 RPM range.'

COMMISSION EXPERIENCES — CLOCKMAKER, CABINETMAKER, SPEAKER BUILDER *e.g. 'The clockmaker (Horological Workshop, Munich) understood the remontoire brief immediately. Lead time was 14 weeks — within the documented range. The first motor's run time was 22 minutes, short of the 25-minute target. After one round of adjustment, second delivery was 26 minutes.' Names shared with permission only.*

SECTION 7

Problems *Encountered*

Problems you encountered that are not in Section XI (Known Problems) of the main document are candidates for addition to the next revision. Describe each problem, how you found it, and how you resolved it (or did not).

PROBLEM 1

Brief title — e.g. 'Bearing hum at 33Hz'

Description, discovery, resolution.

PROBLEM 2

Brief title

PROBLEM 3

Brief title

ADDITIONAL PROBLEMS (CONTINUE ON SEPARATE SHEET IF NEEDED)

SECTION 8

Subjective *Assessment*

Technical measurements tell us what the instrument does. This section tells us what it is like. Future builders will read this before they start — be honest and be specific.

SOUND QUALITY – YOUR HONEST ASSESSMENT

Describe what the instrument actually sounds like in plain language. Not audiophile vocabulary — plain description. 'The bass is physically felt in the stomach at normal listening volume' is useful. 'Exceptional soundstage and PRaT' is not. If it does not sound as described in the document, say so and describe what it actually sounds like.

BUILD EXPERIENCE – WHAT IT IS ACTUALLY LIKE TO BUILD THIS

The hardest part. The most satisfying part. The part that took unexpectedly long. The part that was easier than expected. What would you warn the next builder about? What would you encourage them about?

OPERATING EXPERIENCE – WHAT IT IS LIKE TO USE IN PERFORMANCE

How does the cranking feel in practice? Is the timing of crank cycles manageable during a mix? Did the instrument behave as documented during a real performance? Any surprises?

AUDIENCE AND VENUE REACTION (IF PERFORMED PUBLICLY)

How did audiences respond to the instrument? Did the Option D audience crank unit work theatrically as intended? Any observations about venue suitability?

SECTION 9

Advice for the *Next Builder*

THE SINGLE MOST IMPORTANT THING YOU WOULD TELL THE NEXT BUILDER

One thing. The most important thing. If you could say one sentence to the person who is about to start this build, what would it be?

WHAT TO DO FIRST (BEFORE ANYTHING ELSE)

What should the next builder sort out before they touch a single tool?

WHAT TO DO DIFFERENTLY

If you built this again, what would you change about your process — not the design, but your approach?

SUPPLIER RECOMMENDATIONS (BY REGION IF RELEVANT)

Suppliers who were helpful, responsive, and produced good work. Suppliers to avoid. Regional alternatives that worked well.

SECTION 10

Media & *Attachments*

Photographs, measurements, recordings, and drawings submitted with this report will be hosted at mumblejinx.github.io/opensource with your attribution. All media submitted is assumed to be released under CC BY-NC unless otherwise noted.

- [ ] Build photographs attached — in-progress and completed (label each with the build stage)
- [ ] Measurement data attached — raw figures from calibration tests in CSV or tabular format
- [ ] Audio recording attached — first playback or performance recording (any format)
- [ ] Modified drawings or schematics attached — if you produced improved or corrected versions
- [ ] Video attached — build process or performance (YouTube link acceptable)

LINKS TO EXTERNAL MEDIA (YOUTUBE, INSTAGRAM, PERSONAL SITE, ETC.)

PERMISSION TO USE YOUR REPORT CONTENT IN DOCUMENT REVISIONS

Yes, attributed · Yes, anonymised · No

The project may quote or paraphrase your findings in updated versions of the builder package. Your attribution will match whatever name or handle you provided in Section 1.

HOW TO SUBMIT

Submit this completed report — printed and scanned, or as a filled PDF — to **mumblejinx.github.io/opensource**. If the website is not yet live when you complete your build, send directly to the project contact listed on the main document. Every report received will be acknowledged and its key findings will be incorporated into the next document revision.

The Volta Confluence Engine was designed to be built. You built it. That matters more than any of the details in this form. Thank you.

INSTRUMENT

Volta Confluence Engine
Human-Powered Two-Channel Phonograph Mixer

DOCUMENT

Builder's Report Template
VCE-BRT-001 · Revision 1.0
2025

PROVENANCE

Concept: Jake Galm, researched & documented 2026
Named for Alessandro Volta
No wall connection · No batteries